

- Q.26 What is population forecasting? Explain any two methods of population forecasting. (CO2)
- Q.27 Explain excavation of trench and what are precautions taken during excavation. (CO6)
- Q.28 Explain any five water supply fixtures and their installations. (CO7)
- Q.29 Explain conservancy system give its advantages. (CO8)
- Q.30 Explain construction of drop manhole and give its functions. (CO9)
- Q.31 Write a short note Construction of surface drains. (CO10)
- Q.32 Write a short note explaining C.O.D. (CO11)
- Q.33 Write a short note on nuisance due to sewage disposal. (CO12)
- Q.34 Explain the working of sludgedigestion tank. (CO13)
- Q.35 Draw and explain different types of cisterns. (CO14)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 What is Sedimentation? Give its purpose. Also explain type sedimentation tanks. (CO4)
- Q.37 What is water distribution system? Give its requirements and explain methods used for distributing water. (CO5)
- Q.38 a) Explain primary clarifiers and secondary clarifiers.(5 marks) (CO13)
- b) Write a short note on sludge treatment and its disposal. (5 marks)

Note: Course Outcome (CO) mentioned in the question paper is for official purpose only.

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**4th Semester/ Civil, Brick Tech., Constr. Mgmt.
Highway Engg.,
Subject: Water Supply &
Waste Water Engineering / Public Health Engg.**

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 In which method of population forecasting, increase in population from decade to decade is assumed constant? (CO2)
- Arithmetical increase method
 - Geometrical increase method
 - Incremental increase method
 - Decreased rate of growth method
- Q.2 A coagulant aid is one which (CO4)
- Promotes coagulation
 - Stimulates floc formation
 - Increases dosed of coagulant
 - Narrows down the pH range
- Q.3 Among the following water supply distribution system, which one is referred to as Reticulation system? (Co5)
- Grid Iron system
 - Radial system
 - Dead end system
 - Ring system

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- Q.4 What helps the pipe line to branch off at 90°? (CO6)
 a) Elbows c) Plug
 b) Tee branch d) Coupling
- Q.5 The waste which include grass, furniture, wastepaper, waste fruits is called _____. (CO8)
 a) Dry waste c) Both A & B
 b) Rubbish d) None of these
- Q.6 Which one does not affect the sanitary sewage (CO9)
 a) Population of area c) Plan of area
 b) Rain water d) None of these
- Q.7 Timbering of trenches is necessary for (CO10)
 a) Supporting the sides of trench
 b) When water table rises
 c) Proper joints of sewers
 d) Both A & B
- Q.8 The degree of turbidity can be measured by (CO11)
 a) Turbidity rod c) Both A & B
 b) Turbidimeter d) None of these
- Q.9 At about _____ Moisture, the sludge becomes dry. (CO13)
 a) 10% c) 20%
 b) 15% d) 5%
- Q.10 What is the name of the water coming out of the kitchen, bathroom and wash basin? (CO12)
 a) Garbage c) Sewage
 b) Sullage d) Discharge

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 What is confined aquifer? (CO2)
 Q.12 Explain Coagulants? (CO4)
 Q.13 Define term sanitation. (CO5)
 Q.14 The process of placing the excavated soil back into the trench all round the pipes is called _____. (CO6)
 Q.15 Human excreta is a _____ waste. (CO8)
 Q.16 Masonary sewers are used when the size is _____. (CO9)
 Q.17 Water test is conducted for _____. (CO10)
 Q.18 Aerobic bacteria takes free oxygen in a _____. (CO11)
 Q.19 In primary clarifiers water is detained for _____ hours. (CO12)
 Q.20 Raw sludge contains _____ of moisture content. (CO13)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 What is water meter? Give its advantages and disadvantages. (CO5)
 Q.22 Explain variation in rate of demand. (CO2)
 Q.23 What is Water supply system. (CO1)
 Q.24 Write a short note on maintenance of purity of water. (CO3)
 Q.25 Write a short note on sedimentation with coagulation. (CO4)